Multigrid Self-Gravity in the AthenaK Shearing Box: I. Mesh-Refinement Infrastructure (Phase 0)

`athenak-multigrid` tree, branch `multigrid`

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Abstract

This document opens the *multigrid track* of the self-gravity program: the route to gas + dust self-gravity in AthenaK shearing boxes *with mesh refinement*. It records Phase 0 — the shearing-box/refinement infrastructure every later phase rests on: (i) an `orbital_advection` toggle with a fully consistent non-FARGO (full-velocity) mode; (ii) an enforced *interior-*x_1 refinement policy replacing AthenaK's blanket shearing-box+refinement prohibition; and (iii) orbital advection (FARGO) on refined meshes under an *annular* refinement policy (refined regions span the full azimuthal extent). Validation: a refined-ring run reproduces a reference SMR run bit-for-bit in non-FARGO mode; with FARGO on the ring, mass is conserved exactly and a 20-orbit epicycle test tracks the analytic solution with an amplitude drift of only 0.008% (versus 0.064% on the uniform grid), with the epicyclic amplitude invariant flat to 10^{-4}. The companion document `fft_selfgravity_validation.pdf` covers the FFT solver used as the machine-precision oracle throughout this program; a Julia notebook (`multigrid_validation_tests.ipynb`) reproduces every test here.

Contents

1 Plan and scope

1.1 Goal and roadmap

The scientific target is shearing-box simulations combining gas self-gravity, dust with mutual drag (streaming instability, including the background radial pressure gradient), and mesh refinement that zooms onto high dust-density regions in spiral arms launched by gravitational instability (cf. Baehr, Zhu & Yang 2022, ApJ 933, 100, who omitted the pressure gradient). Mesh refinement is the reason AthenaK was chosen, so the Poisson solver must ultimately live on refined meshes — which an FFT cannot. The adopted route therefore makes the *multigrid* solver shear-aware, with the validated FFT solver retained in three roles: machine-precision oracle for the multigrid shear boundary conditions, production solver for uniform-grid runs, and donor of exact open-z Dirichlet boundary values for stratified multigrid runs.

The phases of the multigrid track:

1. **Phase 0 (this document, done):** shearing box + refinement infrastructure. 0a: an `orbital_advection` on/off toggle with a consistent full-velocity (non-FARGO) mode, and the interior-x_1 refinement policy. 0b: FARGO on refined meshes under the annular refinement policy. Without 0b, refined shearing boxes pay a timestep penalty of $\sim (c_s + q\Omega L_x/2)/c_s \approx 10$–30 for gravito-turbulence boxes.

2. **Phase 1:** shear-periodic boundary conditions in the multigrid driver (`MultigridDriver`), validated against the FFT oracle: exactness at shear-periodic instants, shearing-wave convergence at generic shear phase, and the swing-amplification test with `solver=multigrid`. The V-cycle convergence rate with remapped boundaries is the one open risk of the program and is measured here.

3. **Phase 2:** multigrid gravity on refined shearing-box meshes (SMR first, then AMR), compared against high-resolution uniform FFT references.

4. **Phase 3–5:** merge with the `athenak-dust` particle-mesh dust module (dust density in the Poisson source, gravitational force on particles, pressure-gradient source term for the streaming instability), stratification with open-z boundaries, and the science runs.

1.2 Phase-0 scope and restrictions

What works after Phase 0, all enforced by fatal errors rather than conventions:

- Hydrodynamic shearing boxes with SMR/AMR, provided refined MeshBlocks never touch the shear-periodic x_1 boundaries (keep one root-level block between refinement and the boundary so 2:1 balancing cannot propagate refinement there).

- Orbital advection on refined meshes, provided every refined region spans the full x_2 (azimuthal) extent — *annular rings*, interior in x_1. Without FARGO (`orbital_advection=false`) arbitrary interior refinement shapes are allowed.

- AMR flag handling that keeps rings intact (refine a ring if any member is flagged, derefine only if all are).

Not yet supported/exercised: MHD orbital advection with refinement (the face-centered EMF remap at level interfaces needs a $\nabla \cdot B$ analysis; fatal-guarded), dynamic AMR ring creation/destruction end-to-end (the flag logic is in place; the natural test arrives with the dust-density refinement criterion), and a production-size demonstration of the FARGO speedup itself. SMR/AMR runs must write `bin` outputs — the vtk writer does not support refined meshes.

2 Numerical method

2.1 Two frames for the shearing box

With background shear flow $\mathbf{v}_0 = -q\Omega x\,\hat{y}$, AthenaK can now evolve either of two variable sets, selected by `<shearing_box> orbital_advection`:

FARGO frame (`orbital_advection = true`, default). The stored velocity is the *fluctuation* $\mathbf{v}' = \mathbf{v} - \mathbf{v}_0$; the background advection is applied analytically once per step by the orbital-advection shift (Sec. 2.2). The momentum source terms are

$$\dot{M}_x = 2\Omega\,\rho v'_y, \qquad \dot{M}_y = -(2-q)\,\Omega\,\rho v'_x, \tag{1}$$

where the $(2-q)$ coefficient is the sum of Coriolis (-2Ω) and the advection of background shear by the radial fluctuation $(+q\Omega)$. The CFL condition sees only $|\mathbf{v}'|$.

Full-velocity frame (`orbital_advection = false`). The stored velocity includes the shear. The source terms are the full Coriolis force plus the tidal force, and (for an ideal EOS) the tidal work:

$$\dot{M}_x = 2\Omega\,\rho v_y + 2q\Omega^2 x\,\rho, \qquad \dot{M}_y = -2\Omega\,\rho v_x, \qquad \dot{E} = 2q\Omega^2 x\,\rho v_x \tag{2}$$

(the Coriolis force does no work). Because the wrapped data at the shear-periodic x_1 faces comes from the opposite side of the box, where the background azimuthal velocity differs by $q\Omega L_x$, the boundary exchange must offset the wrapped azimuthal momentum and energy:

$$M_y \to M_y + \rho\,\Delta v, \qquad E \to E + M_y\,\Delta v + \tfrac{1}{2}\rho\,\Delta v^2, \qquad \Delta v = \pm\,q\Omega L_x \tag{3}$$

(+ at the inner face, − at the outer). Initial conditions must include $\mathbf{v}_0$. The two frames are the same physics on different variables; comparing them is a strong consistency test (Sec. 5). The cost of the full-velocity frame is the timestep,

$$\frac{\Delta t_{\mathrm{FARGO}}}{\Delta t_{\mathrm{full}}} \approx \frac{c_s + q\Omega L_x/2}{c_s + |\mathbf{v}'|_{\mathrm{max}}}, \tag{4}$$

measured directly as 5.7× for the $L_x = 2\pi$ swing-test box (Sec. 5.1) and reaching 10–30× for gravito-turbulence boxes — plus the extra truncation diffusion of advecting structures through the grid at many c_s. Non-FARGO is therefore the correctness baseline, not the production mode.

2.2 Orbital advection and its refinement problem

The orbital-advection step shifts every x-column of every MeshBlock by the background displacement $\delta y(x) = q\Omega\,x\,\Delta t$, as an integer number of cells plus a conservative fractional remap,

$$q_j \;\to\; q_{j-j_{\mathrm{off}}} - \left(F_{j+1-j_{\mathrm{off}}} - F_{j-j_{\mathrm{off}}}\right), \qquad j_{\mathrm{off}} = \left\lfloor \frac{\delta y}{\Delta y} \right\rfloor, \tag{5}$$

with upwind remap fluxes F of first (`dc`), second (`plm`), or third (`ppmx`) order — the same kernels as the shear-periodic boundary remap. The required y-ghost data travels in buffers of depth n_g + `maxjshift` exchanged with the two x_2-face neighbors.

The crucial observation for refinement: *the shift kernel is already per-block correct on a multilevel mesh.* Each block computes δy from its own cell centers and applies it in its own Δy; blocks at different levels shift by the same physical displacement expressed in their own cell units, and after the shift the ordinary AMR ghost exchange (with prolongation/restriction at x_1/x_3 level interfaces) restores consistency to interpolation order, since a uniform-in-y translation commutes with the interpolations to that order. What is *not* level-safe is the communication pattern: the buffer exchange assumes both x_2-face neighbors are same-level, same-size blocks, which fails at any coarse–fine interface in y. This is exactly why refinement previously worked only with orbital advection disabled.

2.3 Refinement policies

Rather than generalizing the y-communication to level interfaces, Phase 0 removes the interfaces:

Interior-x_1 policy (always). Refined MeshBlocks must not touch the shear-periodic x_1 boundaries, so the shear-boundary remap only ever involves root-level blocks and the existing machinery applies unchanged. Enforced fatally, block-by-block, at tree build, restart, and after every AMR update (`Mesh::CheckShearingBoxRefinement`); an AMR-side veto keeps boundary blocks from being flagged.

Annular policy (when FARGO is on). Every refined region must span the full x_2 extent over its x_1–x_3 footprint: refinement comes in *rings*. Within a ring every x_2-face neighbor is same-level and same-size, so the per-level shift plus the *existing* communication code are valid with no changes. Enforced by ring-completeness counting over (level, lx_1, lx_3); a partial ring with FARGO on is a fatal error naming the offending ring. For AMR, refine/derefine flags are synchronized ring-wise after the global flag exchange — refine the whole ring if any member is flagged, derefine only if every member is — deterministically on all ranks. Scientifically the ring shape is a good match for the target problem: GI spiral arms are azimuthally extended, and a ring at the arm's radius captures the whole arm and every clump in it.

Buffer-depth safety. The integer shift on a refined block at radius x is $q\Omega|x|\Delta t/\Delta y_\ell$ cells, which the construction-time estimate `maxjshift` $= \lfloor \mathrm{cfl} \cdot q\Omega \max|x| \rfloor + 1$ (now including the previously missing $q\Omega$ factor, doubled on multilevel meshes) may underestimate in unusual configurations. The unpack step therefore re-verifies the actual worst-case shift against the communicated buffer depth every step and aborts with a diagnostic rather than silently corrupting data.

3 What was built

Changes live in commits `e7ef37f2` (infrastructure) and `f1468da3` (bin outputs + docs); Phase 0a is ported from the `athenak` branch `test/shwave-smr`, with provenance kept in the input-file comments.

File (under `src/`)	Change
`hydro/hydro.cpp`, `mhd/mhd.cpp`	`orbital_advection` toggle; MHD+refinement fatal
`shearing_box/shearing_box.{hpp,cpp}`	store the toggle
`shearing_box/shearing_box_srcterms.cpp`	dual-frame source terms, Eqs. (1)–(2)
`shearing_box/shearing_box_cc.cpp`	non-FARGO BC momentum/energy offset, Eq. (3)
`shearing_box/orbital_advection.cpp`	`maxjshift` estimate: $q\Omega$ factor, $2\times$ multilevel headroom
`shearing_box/orbital_advection_cc.cpp`	runtime shift-vs-buffer-depth guard (fatal)
`mesh/mesh.cpp`	blanket prohibition → policy warning
`mesh/build_tree.cpp`	`Mesh::CheckShearingBoxRefinement`: interior-x_1 and ring completeness
`mesh/mesh_refinement.{hpp,cpp}`	boundary-block refinement veto; ring-wise AMR flag sync
`pgen/tests/shwave.cpp`, `swing.cpp`	background shear in ICs when `orbital_advection=false`
`inputs/shearing_box/` (repo root)	`*_smr*.athinput`, `epicycle*.athinput` test inputs (`bin` outputs)

Runtime options (`<shearing_box>` block): `orbital_advection = true|false` (default `true`). Everything else is policy, enforced automatically. The gravity solver selection of the companion document (`<gravity> solver = multigrid|fft`) is unaffected by any of this; Phase 1 will add `shear_periodic` boundary support to the multigrid driver behind the same input block.

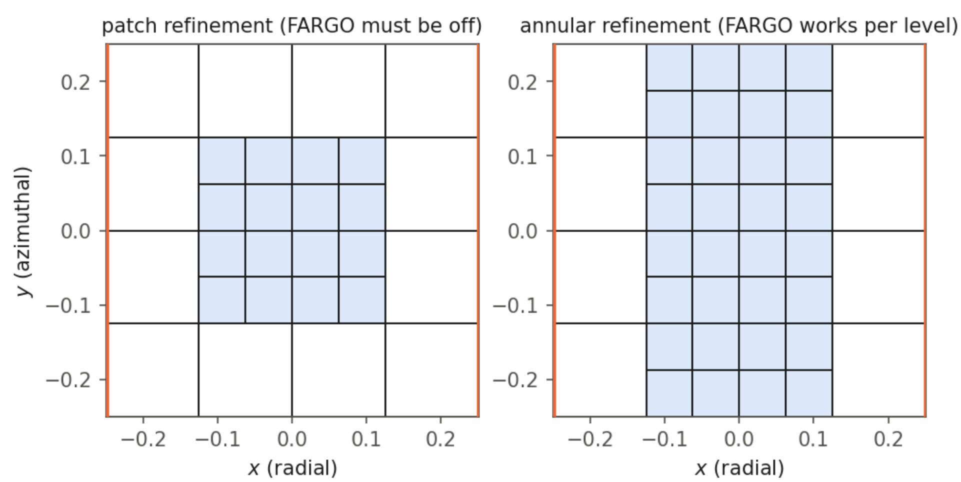

Figure 1: The two refinement layouts, drawn from the `bin` output block structure of actual runs (orange lines: shear-periodic x_1 boundaries; shaded: level-1 blocks). Left: a patch interior in both x_1 and x_2 — legal only with `orbital_advection=false`, because the coarse–fine interfaces in y break the shift communication. Right: the annular layout — the ring has no level jump in y, so FARGO runs per level with unmodified communication.

4 Building and compiling

Phase 0 needs no external dependencies beyond AthenaK's vendored Kokkos:

```
cd athenak-multigrid
git submodule update --init kokkos     # once
```

```
mkdir build && cd build
cmake ..                                # add -D Athena_ENABLE_FFT=ON for the FFT
make -j8                                #   oracle / swing+gravity cross-checks
```

Notes: SMR/AMR runs need `<mesh> nghost` ≥ 4 (even, for prolongation); the FFT flag is only required for the tests that put gravity in the loop (it fetches kokkos-fft and needs `brew install fftw` on macOS, as detailed in the companion document). All test inputs referenced below are under `inputs/shearing_box/` and `inputs/tests/`; runs in this document were executed from `validation/run/` and `validation/run_epicycle/`.

5 Tests and results

5.1 Frame consistency: FARGO vs. full-velocity, uniform grid

```
ATH=build/src/athena
$ATH -i inputs/shearing_box/hydro_incompress_shwave.athinput job/basename=shw_fargo
$ATH -i inputs/shearing_box/hydro_incompress_shwave.athinput job/basename=shw_nofargo \
     shearing_box/orbital_advection=false
# with gravity in the loop (requires FFT build):
$ATH -i inputs/tests/swing_selfgrav.athinput job/basename=SwingNoFargo \
     shearing_box/orbital_advection=false
```

The two frames agree on the frame-independent diagnostics of the JG05 vortical shearing wave — mass exactly, x-kinetic-energy history to 3.3% (the truncation gap between two different discretizations of the same problem). With self-gravity, the non-FARGO swing test tracks the linearized shearing-sheet solution with $L_2(\delta) = 1.2 \times 10^{-2}$ (FARGO: 0.9×10^{-2}) and phase-locking of the density to the instantaneous shearing wavevector at the 10^{-12} level; it costs 5.7× more cycles, exactly Eq. (4).

5.2 Refinement without FARGO (Phase 0a)

```
$ATH -i inputs/shearing_box/hydro_incompress_shwave_smr.athinput \
     shearing_box/orbital_advection=false
```

An interior level-1 patch (Fig. 1, left) in the JG05 vortical-shwave problem. The run reproduces the reference run from the original `test/shwave-smr` branch **bit-for-bit** in every history column — the port changed nothing.

5.3 Guard checks (must abort)

```
# refined region snapping to a shear-boundary block: interior-x1 policy
$ATH -i inputs/shearing_box/hydro_incompress_shwave_smr.athinput \
     refined_region1/x1min=-0.24 refined_region1/x1max=-0.13
# partial-x2 ring with FARGO on: annular policy
$ATH -i inputs/shearing_box/hydro_incompress_shwave_smr.athinput \
     shearing_box/orbital_advection=true
```

Both abort with explanatory fatal errors; the second names the offending ring ("ring at level=3 lx1=2 lx3=0 has 4 of 8 MeshBlocks").

5.4 FARGO on an annular ring (Phase 0b)

```
$ATH -i inputs/shearing_box/hydro_incompress_shwave_smr_ring.athinput \
     job/basename=ring_fargo
$ATH -i inputs/shearing_box/hydro_incompress_shwave_smr_ring.athinput \
     job/basename=ring_nofargo shearing_box/orbital_advection=false
```

The full-x_2 level-1 ring of Fig. 1 (right). With FARGO active on the refined mesh: mass conserved exactly (drift 0.0); x-kinetic-energy history agrees with the non-FARGO twin to 3.8% — the same truncation-level gap the two frames show on uniform grids — and with the uniform-grid FARGO run to 4.1%.

5.5 The endurance test: epicycles for 20 orbits

The `ipert=1` epicycle has spatially uniform $v_x(t)$, $v'_y(t)$: an exact nonlinear solution that any correct y-shift must preserve *exactly*, making it the sharpest probe of the per-level shift and its ring-boundary communication — any indexing or buffer defect appears as $O(1)$ corruption. Two long runs ($L = 20$, $128^2 \times 4$ effective resolution, ideal EOS with $c_s = \sqrt{5/3}$, $v_x(0) = 0.1 = 0.077\, c_s$, 20 orbits = 20 epicyclic periods since $\kappa = \Omega$ at $q = 3/2$):

```
$ATH -i inputs/shearing_box/epicycle.athinput           # uniform 128x128x4
$ATH -i inputs/shearing_box/epicycle_smr_ring.athinput   # level-1 ring + FARGO
```

run	v_x range / amp	amplitude drift, 20 orbits	max $\lvert v_x - \text{analytic}\rvert$ / amp
uniform 128^2	$[-1.0006, +1.0005]$	+0.064%	2.4%
annular ring + FARGO	$[-1.0000, +1.0001]$	+0.008%	0.63%

Both runs hold the ± 1 envelope; the amplitude invariant is flat to a few $\times 10^{-4}$ (uniform) and 10^{-4} (ring) — a slow secular *growth* characteristic of RK2 on an oscillator, not shearing-box or refinement physics. The 3.3% peak difference between the runs is pure phase drift: the ring run takes smaller steps (finer cells) and accumulates *less* RK2 phase error, which is why the refined-mesh FARGO run is the one closer to the analytic solution. A defect in the ring communication would instead show as amplitude noise — none is present at the 10^{-5} level.

5.6 Summary

Check	Metric	Result
SMR no-FARGO vs `test/shwave-smr` reference	all history columns	bit-identical
Uniform: FARGO vs non-FARGO	mass / x-KE	exact / 3.3%
Swing + gravity, non-FARGO vs linear theory	$L_2(\delta)$ / phase	1.2×10^{-2} / 10^{-12}
Non-FARGO cost (L_x=2π box)	cycles vs FARGO	5.7×, as predicted
Ring + FARGO: mass conservation	max drift	exact (0.0)
Shwave: ring-FARGO vs ring-non-FARGO	x-KE	3.8%
Epicycle, 20 orbits, ring + FARGO	amplitude drift / invariant	0.008% / flat to 10^{-4}
Boundary-touching refinement	—	fatal, correct diagnosis
Partial-x_2 ring with FARGO	—	fatal, names the ring

Next: Phase 1 — `shear_periodic` boundary conditions in the multigrid driver, validated against the FFT oracle (exactness at shear-periodic instants; shearing-wave convergence order at generic phase; the swing test with `solver=multigrid`), with the V-cycle convergence rate under remapped boundaries as the quantity to watch.

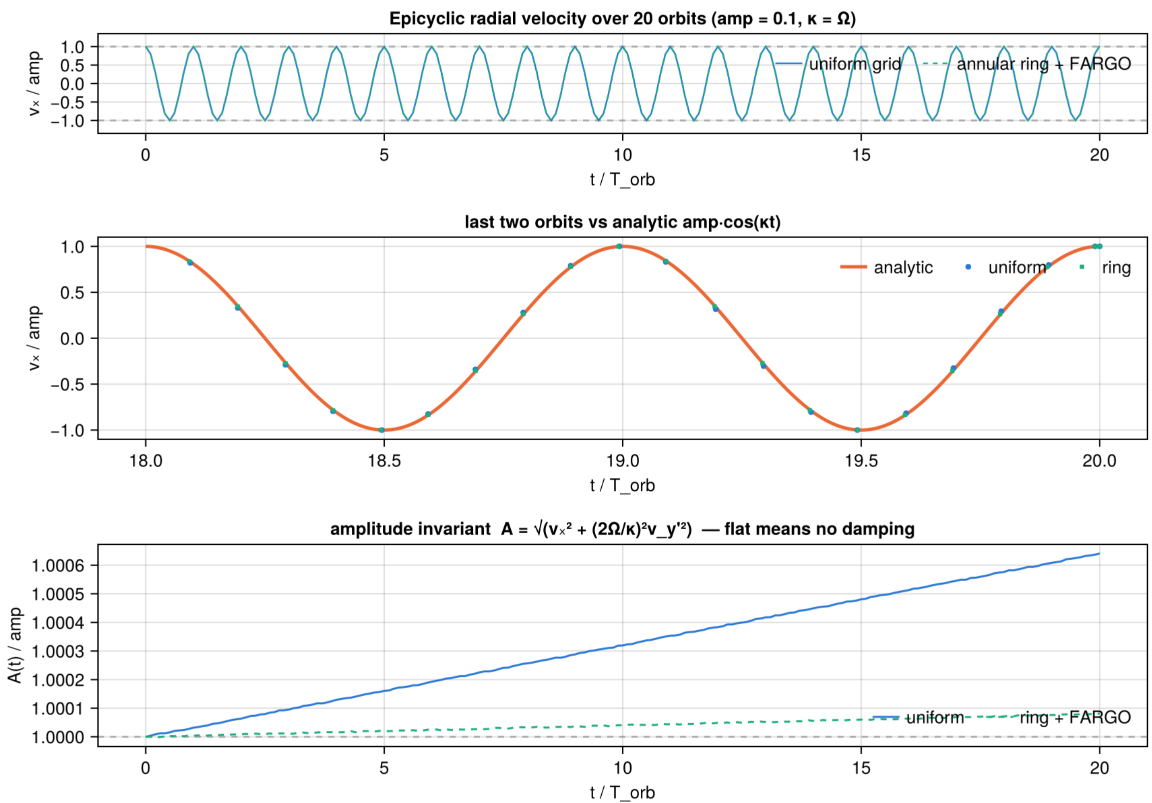

Figure 2: Volume-averaged radial velocity of the epicycle test over 20 orbits, normalized to the initial amplitude. Top: both runs oscillate between ± 1 with no visible decay. Middle: the final two orbits against the analytic $v_x = A\cos\kappa t$. Bottom: the epicyclic amplitude invariant $A = [v_x^2 + (2\Omega/\kappa)^2 v_y'^2]^{1/2}$, which isolates numerical damping/growth from phase error.